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Saturday

ASSESSMENT - Co1

Co1 - AT - B

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IT A1 416

ETHICAL HACKING

- 1) A Company hires you to test its network but gives you only one-line email authorization with no defined scope, half way through testing, you discover a Critical vulnerability in System that wasn't mentioned in email. Do you exploit it to prove impact?

Introduction

Ethical hacking process of legally identifying Security Computer Systems and application permission. Ethical hackers must follow Legal Requirements, professional Ethics and Authorized.

Problem Statement

During the assessment, the Ethical hacker discover a Crucial vulnerability in another System that is not mentioned in Authorization.

Ethical principles involved

Ethical hacking is Governed by Several important principles that ensure testing performed Responsibly.

> Authorization means obtaining official permission before performing testing.

Scope of Engagement

> Scope of Engagement Specifies what Systems Network Application may be tested.

> prevents accidental tests Systems

> Avoid legal disputes

> Ensures Controlled penetration testing.

Why you should Not Exploit the Vulnerability.

Although the Vulnerability Critical, exploiting it would be unethical because.

> Lack of permission

> Business Risk.

> Legal Risks

> Loss of Client

Advantages.

1) Legal protection

2) Improved Client Trust

3) Better Security

4) Reduced operational Risk.

5) professional Reputation.

- 2) You are conducting a penetration test and discover that Client's production database read Customer financial data. You test gates for exploiting a SQL injection vulnerability. How do you proceed without causing harm, and what precaution should have been before engagement began?

Introduction

penetration testing Authorized Security used to organization system attackers exploit them. one of the most common application in SQL injection.

Problem statement

During the penetration test, an Ethical hacker discovers production Real Customer financial information bank account numbers, transaction Records and personal details.

Ethical principles involved

- 1) Authorization
- 2) Confidentiality
- 3) Integrity.
- 4) Availability.
- 5) professional Responsibility.

- 2) You are conducting a penetration test and discover that Client's production database read Customer financial data. You test gates for exploiting a SQL injection vulnerability. How do you proceed without causing harm, and what precaution should have been before engagement began?

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- 3) you discover, during an authorized test, a backdoor that appear to have planted by previous attacker who is still actively using it.

Ans:

Introduction

A penetration test is conducted to identify security weaknesses in organization's system with proper authorization. During such test an ethical hacker may discover.

Under stand the Situation

- > Confirm that the back door genuine and not part of client's software.

> identify how attackers accessing system.

Proper Evidence handling.

- > Collect screenshots, log files, and system information
- > Record the date and time of every observation
- > Maintaining a proper chain of custody for legal purposes

Reporting to the Client

- > Explain the severity of the security breach.
- > Describe how attacker may be using the backdoor.
- > Recommend immediate containment measure

Understanding SQL injection

SQL injection vulnerability where malicious SQL Commands inserted into application to Manipulate

Ex

username : Admin

password : password 123.

An attacker may be

' OR '1' = '1'

Types of SQL injection

- > Error Based SQL injection
- > Union Based SQL injection
- > Blind SQL injection
- > Time Based SQL injection

Advantages of Safe penetration testing

- 1) protects Customer information
- 2) prevent Business Disruption
- 3) Improves Security
- 4) Maintains legal Compliance
- 5) Builds Client Trust.

4) A client terminates your penetration testing contract mid engagement due to business dispute, but you still have active reverse shells and access credentials on their system. Explain, step by step, take immediately.

Introduction
A penetration test is performed only with the client's permission. If client terminates the contract before the test is completed, all authorization immediately.

Stop all testing activities.
Immediately stop scanning, exploitation, and testing.

> Do not access any client system after termination.
> Ensure no further actions performed without permission.

Remove Active Access
Close all reverse shell sessions
> Log out from every client system.

> Remove testing tools or scripts.
> Delete temporarily created accounts for tests.

> Verify that no active connection remains.